

Lecture Notes

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1 Theorem-family boxes

Theorem 1.1 Fundamental Theorem of Calculus

Corollary 1.1 Constant Antiderivatives

If F and G are both antiderivatives of f on an interval I , then $F - G$ is constant on I .

Lemma 1.1 Bounded Monotone Sequences

Every bounded, monotone sequence of real numbers converges.

Claim 1.1 Sum of Two Evens

Let a, b be even integers. Then $a + b$ is even, since $a = 2m$ and $b = 2n$ for some $m, n \in \mathbb{Z}$, so $a + b = 2(m + n)$.

Example 1.1 (Evaluating a Limit)

Compute $\lim_{x \rightarrow 0} \frac{\sin x}{x}$. By the squeeze theorem, this limit equals 1.

Definition 1.1: Continuity

A function f is *continuous* at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Remark 1.1 On Notation

See above for the formal definition of continuity used throughout these notes.

Question 1.1: Open Problem

Does the Collatz conjecture hold for all positive integers?

Note:-

Boxes that share the `cmleftthm` style (theorem, corollary, lemma, claim, remark) are numbered within the section and share the `th` label family; `definition` and `question` use their own `def/qst` prefixes instead.

Python

```
1 def is_prime(n):
2     if n < 2:
3         return False
4     for k in range(2, int(n**0.5) + 1):
5         if n % k == 0:
6             return False
7     return True
```

2 Derivative macros

Ordinary derivative, with and without an explicit function:

- $\backslash dv\{x\}\{f\} \rightarrow \frac{df}{dx}$
- $\backslash dv\{x\} \rightarrow \frac{d}{dx} (x^2 + 3x)$
- $\backslash dv[2]\{x\}\{f\} \rightarrow \frac{d^2 f}{dx^2}$
- $\backslash dv[3]\{x\} \rightarrow \frac{d^3}{dx^3} (\sin x)$

Partial derivative, same four forms:

- $\backslash d dv\{x\}\{f\} \rightarrow \frac{\partial f}{\partial x}$
- $\backslash d dv\{y\} \rightarrow \frac{\partial}{\partial y} (x^2 y + \sin(y))$
- $\backslash d dv[2]\{x\}\{\sin(x) + 7y\} \rightarrow \frac{\partial^2 \sin(x) + 7y}{\partial x^2}$
- $\backslash d dv[2]\{y\} \rightarrow \frac{\partial^2}{\partial y^2} (x^2 y^3)$

A mixed use, e.g. writing out a Laplacian in two variables:

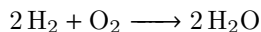
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

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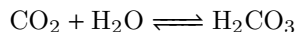
3.1 Basic formulas and reactions

Simple molecular formulas: H_2O , CO_2 , $\text{C}_6\text{H}_{12}\text{O}_6$.

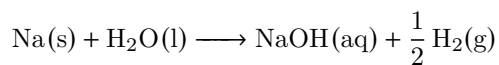
Balanced reaction with an arrow:



Reversible reaction (equilibrium arrows):



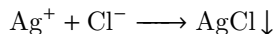
Reaction with states of matter:



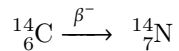
3.2 Charges and isotopes

Ions: Na^+ , Cl^- , SO_4^{2-} , Fe^{3+} .

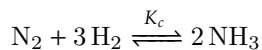
Precipitation reaction:



Isotope notation:



3.3 Equilibrium constant



4 chemfig

4.1 Simple chains

Ethanol, drawn as a chain of bonded groups:

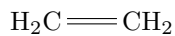


Propane:



4.2 Double and triple bonds

Ethene (double bond):

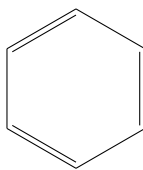


Ethyne (triple bond):

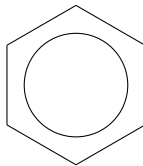


4.3 Rings

Benzene, drawn as a hexagon with alternating double bonds:



Benzene, drawn with the circle:



4.4 Branching

Isobutane, using parentheses to branch off the main chain:

